



LAXMI CHARITABLE TRUST'S
SHETH L.U.J. COLLEGE OF ARTS & SIR M.V. COLLEGE OF
SCIENCE & COMMERCE

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PRACTICAL NO

AIM: Understanding TinyOS Architecture and Execution Model

CODE:

T074 Krishna Chaudhary WSN Pract 2

```
import time
import queue
import threading

class TinyOSSchedulerSimulation:
    def __init__(self):
        self.task_queue = queue.Queue(maxsize=8)
        self.is_running = True
        self.system_lock = threading.Lock()

    def post_task(self, task_name, computation_time):
        try:
            self.task_queue.put_nowait((task_name, computation_time))
            print(f"[SYSTEM]: Task '{task_name}' successfully posted to FIFO queue.")
        except queue.Full:
            print(f"[WARNING]: FIFO Task Queue Full! Task '{task_name}' dropped.")

    def trigger_hardware_event(self, event_name, bound_task_name, comp_time):
        print(f"\n ⚡ [HARDWARE INTERRUPT]: Event '{event_name}' fired!")
        self.post_task(bound_task_name, comp_time)

    def run_scheduler_loop(self):
        print("[INIT]: TinyOS Scheduler Loop Initialized. Waiting for tasks...")

        while self.is_running or not self.task_queue.empty():
            try:
                task_name, comp_time = self.task_queue.get(timeout=1)
```

NAME: KRISHNA NANDKUMAR CHAUDHARY

ROLL NO.: T074

SUBJECT: WIRELESS SENSOR NETWORKS



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```
print(f"\n🔄 [SCHEDULER]: Dispatching '{task_name}' (FIFO Front)...")

with self.system_lock:
    time.sleep(comp_time)
    print(f"✅ [TASK COMPLETE]: '{task_name}' ran to completion in {comp_time}s.")

self.task_queue.task_done()

except queue.Empty:
    if not self.is_running:
        break

print("[POWER]: No pending tasks. MCU entering low-power sleep mode...")
time.sleep(0.5)

if __name__ == "__main__":

    print("T074 Krishna Chaudhary WSN Pract 2\n")

    simulation = TinyOSSchedulerSimulation()

    scheduler_thread = threading.Thread(target=simulation.run_scheduler_loop)
    scheduler_thread.start()

    time.sleep(0.5)

    simulation.trigger_hardware_event("Timer0.fired", "ReadSensorTask", 1.5)
    simulation.trigger_hardware_event("Radio.receive", "ProcessPacketTask", 0.8)

    time.sleep(0.4)

    simulation.trigger_hardware_event("ADC.dataReady", "FilterDataTask", 0.5)

    time.sleep(4.0)
```

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```
simulation.is_running = False
```

```
scheduler_thread.join()
```

```
print("\n[STOP]: Simulation Finished.")
```

OUTPUT:

```
IDLE Shell 3.13.1
Python 3.13.1 (tags/v3.13.1:0671451, Dec 3 2024, 19:06:28) [MSC v.1942 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
===== RESTART: C:/Users/PRIYANKA/Downloads/T074 WSN Practical no. 2.py =====
>>> T074 Krishna Chaudhary WSN Pract 2

[INIT]: TinyOS Scheduler Loop Initialized. Waiting for tasks...

⚡ [HARDWARE INTERRUPT]: Event 'Timer0.fired' fired!
[SYSTEM]: Task 'ReadSensorTask' successfully posted to FIFO queue.
📡 [SCHEDULER]: Dispatching 'ReadSensorTask' (FIFO Front)...

⚡ [HARDWARE INTERRUPT]: Event 'Radio.receive' fired!
[SYSTEM]: Task 'ProcessPacketTask' successfully posted to FIFO queue.

⚡ [HARDWARE INTERRUPT]: Event 'ADC.dataReady' fired!
[SYSTEM]: Task 'FilterDataTask' successfully posted to FIFO queue.
✅ [TASK COMPLETE]: 'ReadSensorTask' ran to completion in 1.5s.

📡 [SCHEDULER]: Dispatching 'ProcessPacketTask' (FIFO Front)...
✅ [TASK COMPLETE]: 'ProcessPacketTask' ran to completion in 0.8s.

📡 [SCHEDULER]: Dispatching 'FilterDataTask' (FIFO Front)...
✅ [TASK COMPLETE]: 'FilterDataTask' ran to completion in 0.5s.
[POWER]: No pending tasks. MCU entering low-power sleep mode...

[STOP]: Simulation Finished.
>>>
```

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